

DEER PARK, WA, USA

WMO: 727870

Lat: 47.974N

Lon: 117.429W

Elev: 2191

StdP: 13.57

Time Zone: -8.00 (NAP)

Period: 01-19

WBAN: 94119

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.9	8.7	-5.8	4.4	4.7	0.6	6.1	12.1	23.7	27.0	19.9	27.9	3.6	340	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.5	92.9	62.3	90.1	61.8	86.4	60.9	65.4	85.2	63.7	84.2	62.1	82.3	6.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.9	80.7	69.5	56.5	73.9	67.1	54.6	68.7	65.9	31.2	85.0	29.9	84.1	28.8	82.4	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	17.2	15.0	-9.1	98.5	8.2	2.9	-15.0	100.6	-19.8	102.3	-24.4	103.9	-30.3	106.1
			-9.6	68.6	8.0	2.7	-15.4	70.5	-20.0	72.1	-24.5	73.6	-30.3	75.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	27.5	30.3	37.6	44.8	53.6	59.3	67.3	66.2	57.4	45.0	34.8	27.4
	DBStd	15.77	8.92	7.75	6.96	6.48	7.12	6.66	5.86	5.65	7.01	7.24	7.79	8.28
	HDD50	3224	697	551	385	183	45	5	0	0	14	189	456	699
	HDD65	7172	1162	971	849	605	358	195	45	52	245	621	905	1164
	CDD50	1775	0	0	1	27	156	284	535	503	235	33	1	0
	CDD65	250	0	0	0	0	4	25	115	90	16	0	0	0
	CDH74	6096	0	0	0	17	206	673	2518	2125	544	13	0	0
	CDH80	2585	0	0	0	2	41	241	1177	941	182	1	0	0
Wind	WSAvg	5.5	4.7	5.3	6.2	6.7	6.4	6.0	5.5	5.2	5.2	5.0	5.1	4.8
Precipitation	PrecAvg	24.5	2.8	1.8	2.9	1.9	1.8	2.0	0.6	0.7	1.0	2.1	3.0	3.4
	PrecMax	31.1	7.7	2.6	4.9	3.0	3.3	4.3	1.5	2.4	2.6	7.3	5.8	5.7
	PrecMin	20.6	1.3	0.6	1.5	1.0	1.2	0.6	0.1	0.1	0.4	0.5	2.0	1.4
	PrecStd	2.9	1.5	0.5	1.0	0.5	0.7	1.1	0.4	0.6	0.5	1.5	1.0	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	51.9	63.3	76.3	84.4	92.6	97.5	97.1	90.7	75.0	57.3	49.1
		MCWB	41.3	43.5	48.5	54.9	59.8	63.7	64.5	62.6	60.8	56.3	51.3	44.8
	2%	DB	42.1	46.5	57.4	69.7	79.6	87.0	93.4	92.6	85.3	68.1	53.1	43.6
		MCWB	38.8	40.3	46.3	52.1	57.9	62.2	62.7	61.7	59.7	53.0	47.7	40.5
	5%	DB	39.2	43.8	53.9	63.6	75.0	81.4	90.5	89.6	80.8	63.1	49.3	39.4
		MCWB	37.0	38.7	44.2	48.9	56.2	60.4	62.2	61.0	57.7	50.8	45.1	37.3
	10%	DB	37.1	40.6	49.7	58.9	70.3	76.9	87.4	85.4	75.2	58.4	46.2	36.9
		MCWB	35.5	36.9	42.1	46.5	53.9	58.0	61.5	60.1	55.6	48.5	42.8	35.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	45.2	50.8	56.8	62.1	66.7	69.4	65.6	63.9	58.7	52.6	45.8
		MCDB	44.6	49.7	58.6	73.6	79.2	86.6	87.3	87.4	81.9	70.0	55.3	47.7
	2%	WB	39.2	42.2	47.8	52.8	59.5	63.8	66.4	63.9	60.6	55.0	48.5	41.0
		MCDB	41.4	45.4	55.7	66.0	75.1	82.2	85.4	86.1	80.3	65.2	51.9	43.6
	5%	WB	37.5	39.5	45.3	50.2	57.5	61.5	64.3	62.4	58.8	52.6	46.0	38.0
		MCDB	39.0	42.7	51.6	62.1	71.9	77.9	84.6	84.3	77.5	60.5	48.5	39.3
	10%	WB	36.0	37.3	43.3	47.9	55.4	59.4	62.6	61.2	56.8	50.1	43.6	35.8
		MCDB	37.2	40.2	49.1	57.2	67.2	74.5	83.2	82.3	73.3	57.4	46.3	36.8
Mean Daily Temperature Range	5% DB	MDBR	11.4	15.7	19.6	23.7	26.3	27.3	34.5	34.6	31.4	23.7	14.7	11.1
		MCDBR	11.3	18.5	26.7	31.5	34.4	36.4	40.6	41.1	39.9	31.6	18.1	12.2
		MCWBR	9.1	12.6	16.7	16.9	15.7	15.5	15.8	16.0	17.9	18.2	13.0	9.9
	5% WB	MCDBR	10.2	15.9	22.7	28.5	30.7	33.2	36.1	35.3	35.0	26.2	15.4	11.2
MCWBR		8.5	11.3	15.1	16.0	14.5	14.9	14.8	14.4	16.1	15.6	11.9	9.4	
Clear-Sky Solar Irradiance	taub	0.278	0.284	0.298	0.322	0.336	0.335	0.335	0.347	0.323	0.305	0.288	0.273	
	taud	2.472	2.511	2.550	2.472	2.454	2.487	2.491	2.453	2.508	2.552	2.545	2.486	
	Ebn at Noon	259	283	296	296	294	294	292	284	282	271	253	246	
	Edh at Noon	20	24	27	32	34	33	33	32	28	23	18	17	
All-Sky Solar Radiation	RadAvg	316	611	935	1409	1815	1935	2307	1917	1361	798	373	251	
	RadStd	44	83	115	136	145	177	141	132	134	95	44	34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+79

Nomenclature: See separate page